

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

Fill out the introduction.

What is the purpose of this section? To introduce background information about phenotyping in relation to plants. - Why it is useful? - Relations to current study at hand?

Context for plant phenotyping? It is of vital importance in plant breeding to be able to propagate desirable heritable traits from plants through controlled human action [8]. This is especially important in the context of food security where a cultivars resistant to adverse abiotic and biotic stresses affect the final overall crop yield. The literature indicates that plant phenotyping has become an effective tool in plant breeding methods. [1].

Points about plant phenotyping Plant phenotyping is then the evaluation of phenotypes in plants with the main goal of producing quantitative data to study the dynamic interaction between a plant and its environment [3].

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [5]. Who stated that a phenotype is all types of organism which are identified from one another by inspection or by some measurement. So it is an observable primitive trait or a complex trait of a plant that can be identified visually or measured and evaluated numerically. Therefore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various different growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant phenotyping suffers from scalability issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogenesis of a cultivar. However in the past decade image processing techniques have been applied successfully to acquire phenotypical data in a non-destructive manner [2]. There exists many different image techniques that have been applied in greenhouses and in the field which allow for the assessment of resistance to disease, biomass, soil salinity, etc using various different sensors, and techniques.

Definition 2: The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [4]

Definition 3: For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture,

and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [3]

Definition 4: The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [7]

Definition 5: Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [6]

Fill out the Image processing techniques section.

Fill out the applications in low cost greenhousing techniques secton.

Fill out the conclusion section.

References

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